

Documentação Técnica - Grafana e Prometheus

1. Visão Geral do Projeto

Este projeto tem como objetivo implementar um ambiente de monitoramento utilizando Prometheus para coletar métricas de uma aplicação hospedada em uma VM na AWS. O ambiente é composto por duas instâncias EC2 e configurações específicas de rede, segurança e exportação de métricas.

2. Arquitetura da Solução

2.1 VMs

- **Monitoramento**
 - Sistema Operacional: Ubuntu 22.04
 - Função: Servidor de monitoramento
 - Ferramentas instaladas: Prometheus
- **Python-EC2**
 - Sistema Operacional: Ubuntu 22.04
 - Função: Servir aplicação a ser monitorada
 - Aplicação em execução: Flask App

3. Recursos da AWS Utilizados

3.1 EC2 Instances

Nome da VM	AMI utilizada	Tipo da Instância	IP Privado	IP Público	Finalidade
monitoramento	Ubuntu 22.04 LTS	t3.micro	...	...	Prometheus
python-EC2	Ubuntu 22.04 LTS	t3.micro	...	...	Aplicação

3.2 VPC

- **CIDR da VPC:** 10.0.0.0/16
- **Sub-redes:** Pública
- **Internet Gateway:** Ativado, para Prometheus acessar recursos externos

3.3 Security Groups

- **SG-prometheus**
 - Porta 9090 liberada para seu IP (acesso ao Prometheus)
 - Porta 22 (SSH)
- **SG-app**
 - Porta 8080 (ou conforme a porta da aplicação)
 - Porta 22 (SSH)

4. Configuração do Prometheus

4.1 Arquivo `prometheus.yml`

```
global:
  scrape_interval: 10s
  scrape_timeout: 10s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'cards_app'
    static_configs:
      - targets: ['44.210.141.191:8080']
```

4.2 Configuração do `App.py`

```
import requests
import time
from flask import Flask
from prometheus_flask_exporter import PrometheusMetrics

app = Flask(__name__)
metrics = PrometheusMetrics(app)

@app.route("/", methods=["GET"])
def index():
    return cards()

@app.route("/cards", methods=["GET"])
@metrics.counter("cards_requested", "Numero de cards solicitados")
def cards():
    response =
requests.get("https://www.deckofcardsapi.com/api/deck/new/shuffle/?deck_count=1")
# https://www.deckofcardsapi.com/api/deck/new/shuffle/?deck_count=1
    cartas = response.json()
    deck_id = cartas["deck_id"]

    image_list = []
    for i in range(0, 14):
```

```

    req =
requests.get(f"https://www.deckofcardsapi.com/api/deck/{deck_id}/draw/?count=2") #
https://www.deckofcardsapi.com/api/deck/{deck_id}/draw/?count=2
    png = req.json()
    image = png["cards"][0]["image"]
    image_list.append(image) # Index out of range

html_template = f"""
    
    
    
    
    
    
    
    
    
    """

    return html_template

@app.route("/latency/<int:ms>", methods=["GET"])
@metrics.gauge("latency", "Latencia em milisegundos (simulacao)")
def latency(ms: int):
    print(f"aguardando: {ms}")
    time.sleep(ms)
    return dict(message=f"Essa página está aguardando {ms} segundos")

@app.route("/404", methods=["GET"])
def http_status_code_404():
    response = requests.get("http://www.google.com")
    return dict(message="ERROR", status_code=response.status_code), 404

@app.route("/health", methods=["GET"])
def health():
    response = requests.get("http://www.google.com")
    return dict(message="SUCCESS", status_code=response.status_code)

app.run(host="0.0.0.0", port="8080")

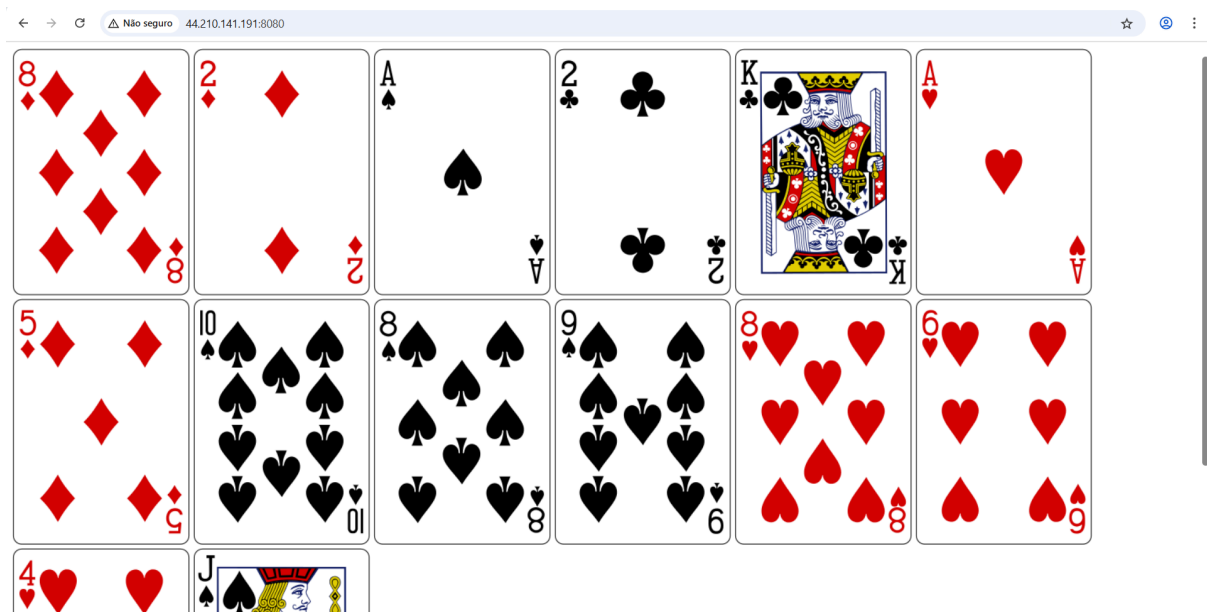
```

5. Métricas Monitoradas

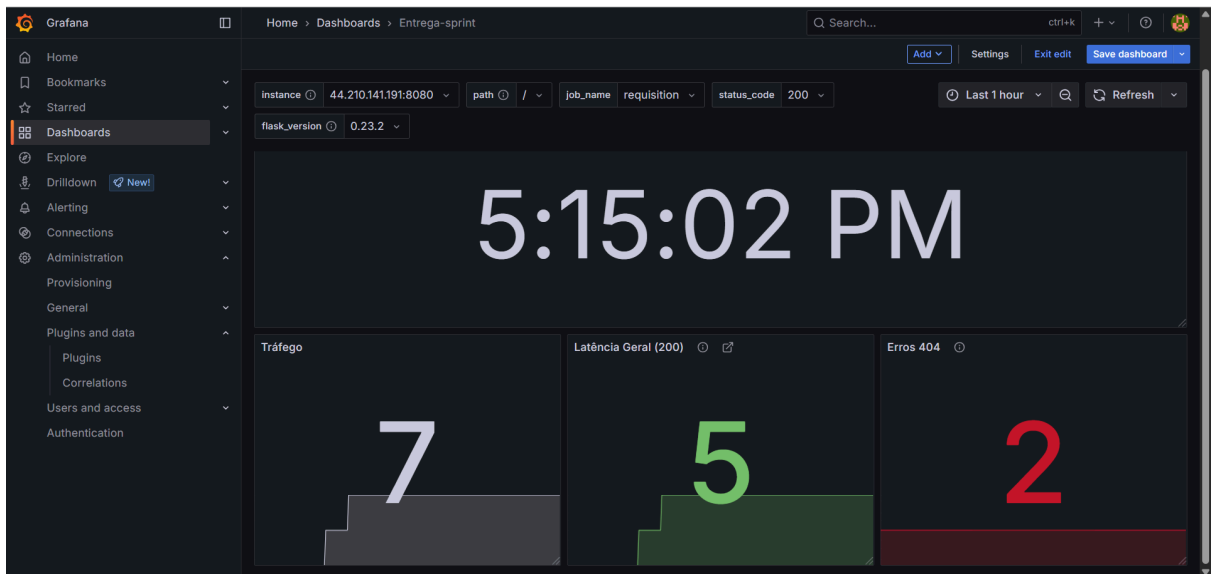
A aplicação expõe as seguintes métricas, coletadas via Prometheus:

1. **instance** - Coleta o IP da instância
2. **path** - Coleta o local onde as métricas estão (/)
3. **job_name** - Mostra o nome do job monitorado
4. **status_code** - Mostra o status que se encontra o site
5. **flask_version** - Mostra a versão do flask que está sendo utilizada no site

6. Prints



(Aplicação rodando na internet)



(Dashboard feito no Grafana)

The image shows the 'Settings' page for the 'Entrega-sprint' dashboard, specifically the 'Variables' tab. The page has a navigation bar with tabs for 'General', 'Annotations', 'Variables', 'Links', 'Versions', 'Permissions', and 'JSON Model'. The 'Variables' tab is active, showing a table of variables. Below the table, there is a '+ New variable' button and a section for 'Renamed or missing variables'.

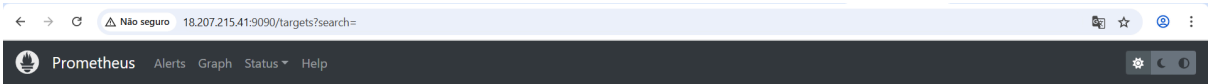
Variable	Definition	
instance	label_values(flask_exporter_info,instance)	
path	label_values(flask_http_request_duration_seconds_count,path)	
job_name	label_values(flask_exporter_info,job)	
status_code	label_values(flask_http_request_total,status)	
flask_version	label_values(flask_exporter_info,version)	

(Métricas utilizadas no Dashboard)

```
44.210.141.191:8080/metrics

# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 2933.0
python_gc_objects_collected_total{generation="1"} 5.0
python_gc_objects_collected_total{generation="2"} 0.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 85.0
python_gc_collections_total{generation="1"} 7.0
python_gc_collections_total{generation="2"} 0.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="12",patchlevel="3",version="3.12.3"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 2.75099648e+08
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 4.585472e+07
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.74958223977e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 5.6000000000000005
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 6.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1024.0
# HELP flask_exporter_info Information about the Prometheus Flask exporter
# TYPE flask_exporter_info gauge
flask_exporter_info{version="0.23.2"} 1.0
# HELP flask_http_request_duration_seconds Flask HTTP request duration in seconds
# TYPE flask_http_request_duration_seconds histogram
flask_http_request_duration_seconds_bucket{le="0.005",method="GET",path="/",status="200"} 0.0
flask_http_request_duration_seconds_bucket{le="0.01",method="GET",path="/",status="200"} 0.0
flask_http_request_duration_seconds_bucket{le="0.025",method="GET",path="/",status="200"} 0.0
flask_http_request_duration_seconds_bucket{le="0.05",method="GET",path="/",status="200"} 0.0
flask_http_request_duration_seconds_bucket{le="0.075",method="GET",path="/",status="200"} 0.0
flask_http_request_duration_seconds_bucket{le="0.1",method="GET",path="/",status="200"} 0.0
```

(Métricas da aplicação)



Targets

All scrape pools ▾

AllUnhealthyCollapse All

Filter by endpoint or labels

Unknown

Unhealthy

Healthy

cards_app (1/1 up)

show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://44.210.141.191:8080/metrics	UP	instance="44.210.141.191:8080" job="cards_app" ▾	6.356s ago	4.236ms	

prometheus (1/1 up)

show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus" ▾	6.596s ago	5.211ms	

(Targets no Prometheus)

```
18.207.215.41 - - [10/Jun/2025 20:11:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:11:15] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:11:30] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:11:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:12:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:12:15] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:12:30] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:12:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:13:00] "GET /metrics HTTP/1.1" 200 -
18.239.24.42 - - [10/Jun/2025 20:13:14] "GET / HTTP/1.0" 200 -
18.207.215.41 - - [10/Jun/2025 20:13:15] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:13:30] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:13:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:14:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:14:15] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:14:30] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:14:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:15:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:15:15] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:15:30] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:15:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:16:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:16:15] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:16:30] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:16:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:17:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:17:15] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:17:30] "GET /metrics HTTP/1.1" 200 -
177.162.82.25 - - [10/Jun/2025 20:17:38] "GET /metrics HTTP/1.1" 200 -
177.162.82.25 - - [10/Jun/2025 20:17:44] "GET / HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:17:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:18:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:18:15] "GET /metrics HTTP/1.1" 200 -
17.152.92.25 - - [10/Jun/2025 20:18:21] "GET / HTTP/1.1" 404 -
18.207.215.41 - - [10/Jun/2025 20:18:30] "GET /metrics HTTP/1.1" 200 -
177.162.82.25 - - [10/Jun/2025 20:18:33] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:18:45] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:19:00] "GET /metrics HTTP/1.1" 200 -
18.207.215.41 - - [10/Jun/2025 20:19:15] "GET /metrics HTTP/1.1" 200 -
]
```

(Aplicação rodando no servidor)

```
root@ip-10-0-0-82:/home/ubuntu# systemctl status prometheus
* prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-06-10 19:15:50 UTC; 1h 4min ago
   Main PID: 10137 (prometheus)
     Tasks: 8 (limit: 1125)
    Memory: 34.4M (peak: 44.8M)
       CPU: 2.710s
   OGroup: /system.slice/prometheus.service
          L:10137 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prometheus/

Jun 10 19:15:50 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:50.730Z caller=main.go:1172 level=info msg="TSDB started"
Jun 10 19:15:50 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:50.730Z caller=main.go:1354 level=info msg="Loading configuration file" filename=/etc/p
Jun 10 19:15:50 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:50.735Z caller=main.go:1391 level=info msg="updated GOOC" old=100 new=75
Jun 10 19:15:50 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:50.735Z caller=main.go:1402 level=info msg="Completed loading of configuration file" filename=/etc
Jun 10 19:15:50 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:50.735Z caller=main.go:1131 level=info msg="Server is ready to receive web requests."
Jun 10 19:15:50 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:50.735Z caller=manager.go:164 level=info component="rule manager" msg="Starting rule manager..."
Jun 10 19:15:59 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:59.958Z caller=compact.go:567 level=info component=tsdb msg="write block resulted in empty block"
Jun 10 19:15:59 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:59.961Z caller=head.go:1335 level=info component=tsdb msg="Head GC completed" caller=truncateMemor
Jun 10 19:15:59 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:59.961Z caller=checkpoint.go:101 level=info component=tsdb msg="Creating checkpoint" from segment
Jun 10 19:15:59 ip-10-0-0-82 prometheus[10137]: ts=2025-06-10T19:15:59.972Z caller=head.go:1317 level=info component=tsdb msg="WAL checkpoint complete" first=27 last=
lines 1-20/20 (200)

I-0be03ac6f52a22d4d (monitoramento)
PublicIPs: 18.207.215.41 PrivateIPs: 10.0.0.82

CloudShell Feedback
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```

(Status do Prometheus)